❤️ Heart Disease Prediction System

📖 Project Overview

The Heart Disease Prediction System is a machine learning–based application that predicts whether a person is likely to have heart disease based on medical parameters.
The system uses a Logistic Regression model trained on a heart disease dataset and provides predictions through an interactive Streamlit web interface.

🎯 Objectives

- Predict heart disease accurately using machine learning
- Provide a user-friendly interface for medical data input
- Assist in early detection and awareness
- Demonstrate end-to-end ML project implementation.

🤖 Machine Learning Model

- Algorithm Used: Logistic Regression
- Type: Binary Classification
- Target Variable:
  - 0 → No Heart Disease
  - 1 → Heart Disease

```
[32]  ✓ 0.1s
model=LogisticRegression(max_iter=10000)
model.fit(x_train,y_train)

...  LogisticRegression
LogisticRegression(max_iter=10000)

[34]  ✓ 0.0s
y_predict=model.predict(x_test)
print(y_predict)

...  [0 1 1 0 1 1 1 0 0 1 1 0 1 0 1 1 1 0 0 0 1 0 0 1 1 1 1 1 0 1 0 0 0 0 1 0 1
 1 1 1 1 1 1 1 1 0 1 1 0 0 0 0 1 1 0 0 0 1 0 0 0]
```

📊 Dataset Information

- Source: Heart Disease Dataset
- Total Records: 271
- Features Used:
  - Age
  - Sex
  - Chest Pain Type
  - Resting Blood Pressure
  - Cholesterol
  - Fasting Blood Sugar
  - Resting ECG
  - Max Heart Rate
  - Exercise Induced Angina
  - Oldpeak
  - Slope
  - Number of Major Vessels
  - Thalassemia

⚙ Technologies Used

- Python
- Pandas, NumPy
- Scikit-learn
- Streamlit
- Pickle

📱 Frontend

- Built using Streamlit
- Interactive input fields with medical explanations
- Real-time prediction output
- User-friendly UI for non-technical users

📈 Model Performance

- Accuracy: ~88.5%
- Precision: ~0.89
- Recall: ~0.88
- F1-score: ~0.88
- Balanced dataset (no class imbalance)

```
accuracy_val=accuracy_score(y_test,y_predict)*100
print(accuracy_val)
[37] ✓ 0.0s
88.52459016393442
```

```
from sklearn.metrics import classification_report, confusion_matrix

print(confusion_matrix(y_test, y_predict))
print(classification_report(y_test, y_predict))
[39] ✓ 0.0s
[[25  4]
 [ 3 29]]
              precision    recall  f1-score   support

           0       0.89      0.86      0.88        29
           1       0.88      0.91      0.89        32

    accuracy                           0.89        61
   macro avg       0.89      0.88      0.88        61
weighted avg       0.89      0.89      0.89        61
```

```
precision_val=precision_score(y_test,y_predict)*100
print(precision_val)
[38] ✓ 0.0s
87.87878787878788
```

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